MATH 152 **QUIZ 3** NAME Solutions

Mrs. Bonny Tighe 25 points

6.5, 7.1 TA_____ Wed 9/18/2019

1. Find the average value of $f(x) = \tan^2(2x)$ *on* $[0, \pi/6]$

$$\frac{1}{\pi/6 - 0}\int_0^{\pi/6} (\sec^2 2x - 1)\,dx = \frac{6}{\pi}\left[\frac{1}{2}\tan 2x - x\right]_0^{\pi/6}$$

$$\frac{3}{\pi}\left[\tan \pi/3 - \pi/3 - 0\right] = \frac{3}{\pi}\left[\sqrt{3} - \pi/3\right]$$

2. The world population in the second half of the 20th century is modeled by the equation $P(t) = 2560e^{0.02t}$ million people, where t is measured in years since 1950. Find the average population during this time.

$$\frac{1}{50}\int_0^{50} 2560e^{0.02t}\,dt = \frac{2560}{50}\frac{e^{.02t}}{.02}\Big]_0^{50} = \frac{2560}{1}e^{.02t}\Big]_0^{50}$$

$$2560\left[e^1 - e^0\right] = 2560\left[e - 1\right] \text{ million people}$$

3. Evaluate the integrals. a) $\int t^3 \ln(t)\,dt$

$u = \ln t \quad dv = t^3 dt$

$du = \frac{1}{t}dt \quad v = \frac{1}{4}t^4$

$$\frac{1}{4}t^4 \ln t - \int \frac{1}{4}t^4\left(\frac{1}{t}\right)dt$$

$$\frac{1}{4}t^4 \ln t - \frac{1}{4}\int t^3\,dt$$

$$\frac{1}{4}t^4 \ln t - \frac{1}{16}t^4 + C$$

$u = \sin 3x \qquad dv = e^{2x}dx$
$du = 3\cos 3x\,dx \qquad v = \frac{1}{2}e^{2x}$

b) $\int e^{2x}\sin(3x)\,dx$

$= \frac{1}{2}e^{2x}\sin 3x - \int \frac{1}{2}e^{2x}\,3\cos 3x\,dx$

$u = \cos 3x \quad dv = e^{2x}dx$
$du = -3\sin 3x\,dx \quad v = \frac{1}{2}e^{2x}$

$= \frac{1}{2}e^{2x}\sin 3x - \frac{3}{2}\int e^{2x}\cos 3x\,dx$

$= \frac{1}{2}e^{2x}\sin 3x - 3/2\left[-\frac{1}{2}e^{2x}\cos 3x - \int \frac{1}{2}e^{2x}(-3\sin 3x)\,dx\right.$

$= \frac{1}{2}e^{2x}\sin 3x + \frac{3}{4}e^{2x}\cos 3x - \frac{9}{4}\int e^{2x}\sin 3x\,dx$

$1 + 9/4 = +13/4$

$= \frac{4}{13}\left[\frac{1}{2}e^{2x}\sin 3x + \frac{3}{4}e^{2x}\cos 3x\right] + C$

$= \frac{2}{13}e^{2x}\sin 3x + \frac{3}{13}e^{2x}\cos 3x + C$

c) $\int \cos\sqrt{x}\,dx \qquad m = \sqrt{x} \quad dm = \frac{1}{2\sqrt{x}}dx \quad dx = 2m\,dm$

$u = 2m \qquad dv = \cos m\,dm$
$du = 2dm \qquad v = \sin m$

$\int 2m\cos m\,dm$

$2m\sin m - \int \sin m\,(2dm)$

$2m\sin m - 2(-\cos m) + C$

$2\sqrt{x}\sin\sqrt{x} + 2\cos\sqrt{x} + C$

d) $\int x^4 \ln x\,dx$

$u = \ln x \qquad dv = x^4\,dx$
$du = \frac{1}{x}dx \qquad v = \frac{1}{5}x^5$

$\frac{1}{5}x^5\ln x - \int \frac{1}{5}x^5\left(\frac{1}{x}\right)dx$

$\frac{1}{5}x^5\ln x - \frac{1}{5}\int x^4\,dx$

$\frac{1}{5}x^5\ln x - \frac{1}{25}x^5 + C$

MATH 152
Mrs. Bonny Tighe

QUIZ 3A

25 points
6.5, 7.1

Name Solution

TA_____ Wed 9/18/2019

1. Find the average value of $f(x)=\dfrac{3^x}{1+3^{2x}}$ *on* $[0,1]$

$u = 3^x$

$du = 3^x \ln 3\, dx$

$\dfrac{du}{3^x \ln 3} = dx$

$$\frac{1}{1-0}\int_0^1 \frac{3^x}{1+3^{2x}}\,dx = \int_1^3 \frac{1}{1+u^2}\frac{1}{\ln 3}\,du$$

$$\frac{1}{\ln 3}\tan^{-1}u\Big]_1^3 \quad \frac{1}{\ln 3}\left[\tan^{-1}3 - \tan^{-1}1\right]$$

$$\frac{1}{\ln 3}\left[\tan^{-1}3 - \pi/4\right]$$

2. A cup of coffee has temperature 95° C and takes 30 minutes to cool to 61° C in a room with temperature 20° C. By Newton's Law of Cooling, the temperature of the coffee after t minutes is $T(t) = 20 + 75e^{-0.5t}$. What is the average temperature of the coffee during the first half hour?

$$\frac{1}{30}\int_0^{30}(20+75e^{-0.5t})\,dt = \frac{1}{30}\left[20t + \frac{75}{-0.5}e^{-0.5t}\right]_0^{30}$$

$$\frac{1}{30}\left[(600-150e^{-.5(30)}) - (-150e^0)\right]$$

$$\frac{1}{30}\left[600-150e^{-15}+150\right]\ {}^oC = \frac{1}{30}\left[750-150e^{-15}\right]\ {}^oC$$

3. Evaluate the integrals. a) $\int \tan^{-1}t\ dt$

$u = \tan^{-1}t \quad dv = dt$

$du = \frac{1}{1+t^2}dt \quad v = t$

$$t\tan^{-1}t - \int \frac{t}{1+t^2}\,dt$$

$$t\tan^{-1}t - \frac{1}{2}\ln|1+t^2| + C$$

b) $\int e^{3x}\cos(2x)\,dx$

$$\boxed{\begin{array}{ll} u = \cos 2x & dv = e^{3x}dx \\ du = -2\sin 2x\,dx & v = \frac{1}{3}e^{3x} \end{array}}$$

$$\frac{1}{3}e^{3x}\cos 2x - \int \frac{1}{3}e^{3x}(-2\sin 2x\,dx)$$

$$\frac{1}{3}e^{3x}\cos 2x + \frac{2}{3}\int e^{3x}\sin 2x\,dx$$

$$\boxed{\begin{array}{ll} u = \sin 2x & dv = e^{3x}dx \\ du = 2\cos 2x\,dx & v = \frac{1}{3}e^{3x} \end{array}}$$

$$\frac{1}{3}e^{3x}\cos 2x + \frac{2}{3}\left[\frac{1}{3}e^{3x}\sin 2x - \int \frac{1}{3}e^{3x}\,2\cos 2x\,dx\right]$$

$$= \frac{1}{3}e^{3x}\cos 2x + \frac{2}{9}e^{3x}\sin 2x - \frac{4}{9}\int e^{3x}\cos 2x\,dx$$

$$+\frac{4}{9}$$

$$\frac{13}{9}$$

$$= \frac{9}{13}\left[\frac{1}{3}e^{3x}\cos 2x + \frac{2}{9}e^{3x}\sin 2x\right] + C$$

$$\frac{3}{13}e^{3x}\cos 2x + \frac{2}{13}e^{3x}\sin 2x + C$$

c) $\int \sin\sqrt{x}\,dx$ $\quad m = \sqrt{x} \quad dm = \frac{1}{2\sqrt{x}}dx \quad dx = 2m\,dm$

$$\int 2m\sin m\,dm$$

$$\boxed{\begin{array}{ll} u = 2m & dv = \sin m\,du \\ du = 2du & v = -\cos m \end{array}}$$

$$-2m\cos m - \int -\cos m\,2du = -2m\cos m + 2\sin m + c$$

$$-2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x} + C$$

d) $\int \sin^{-1} x\,dx$

$$\boxed{\begin{array}{ll} u = \sin^{-1}x & dv = dx \\ du = \frac{1}{\sqrt{1-x^2}}dx & v = x \end{array}}$$

$$x\sin^{-1}x - \int x\frac{1}{\sqrt{1-x^2}}dx = x\sin^{-1}x - \int (1-x^2)^{-1/2}x\,dx$$

$$x\sin^{-1}x + \left(\frac{-1}{2}\right)\cdot 1/\tfrac{1}{2}(1-x^2)^{1/2} + C$$

$$x\sin^{-1}x + \sqrt{1-x^2} + C$$